COURSE INTRODUCTION

BCL1333 SYSTEM FUNDAMENTALS

Overview

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SYNOPSIS

This course introduces the fundamental concepts of computer systems and how hardware and software work together to execute and manage computing tasks. Students will learn the basic components of computer systems, digital logic and CPU organization, including instruction execution, registers, memory hierarchy, and processor performance. The course also introduces parallel and multicore computing, virtualization, system-level programming, error handling, debugging, processes, threads, synchronization, and performance measurement. Through these topics, students develop an understanding of how a computer system operates internally, how hardware and software interact, how programs are executed and managed, and how computing systems can be designed, debugged and measured for reliable and efficient performance.



Learning Outcomes

CLO 1:

Explain fundamental principles of computer systems, including logic design, processor organization, memory hierarchy, parallel processing and virtualization.. (C2, PLO1)

CLO 2:

Analyze sequential and parallel computing solutions using appropriate techniques and tools to determine their reliability and performance.(C4, PLO2)

CLO 3:

Develop system-level solutions incorporating error handling, debugging, synchronization and performance measurement using appropriate computing tools.(P4, PLO3)

Topic to be covered



1. Introduction to Computer Systems
2. Logic Design and Digital Systems
3. Processor Organization and Performance
4. Memory Hierarchy and Storage Systems
5. Parallel Computing and Multicore Systems
6. Virtualization and System Performance
7. System-Level Programming: Error Handling and Debugging
8. Synchronization and Performance Measurement

Tools Used

Tool	Main Chapters	Purpose
IntelliJ IDEA + JDK 21 LTS or later	Ch 3, 5, 7, 8	Write, compile, run, debug, and test Java programs, including parallel programming, error handling, and synchronization
Windows Task Manager	Ch 1, 3, 4, 5, 8	Monitor CPU, memory, processes, threads, and system resource utilisation during program execution
Excel / Google Sheets	Ch 5, 6, 8	Record performance data, calculate execution time, speedup and efficiency, and create charts for analysis
VirtualBox	Ch 6	Create and run virtual machines and compare system performance in physical and virtual environments
Screen Recording Software	Ch 7, 8 / Mini Project	Record debugging activities, code demonstrations, and the student's project walkthrough
LMS	All assessments	Access learning materials, complete online tests, submit assignments and projects, and complete online examinations

Assessment structures

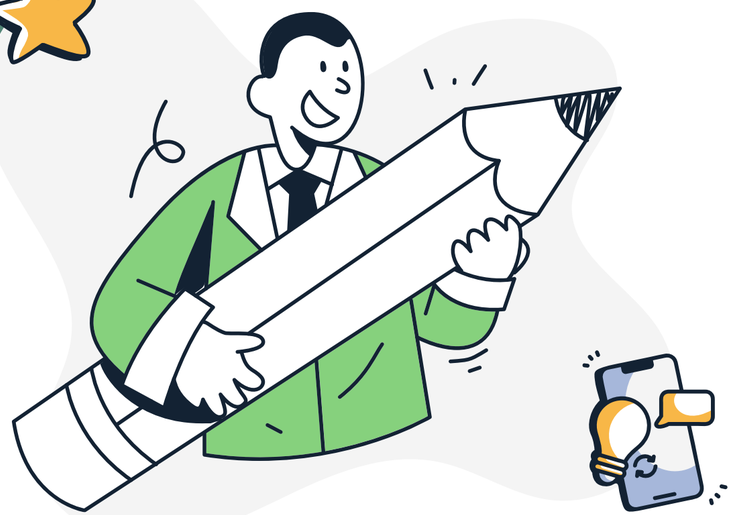
Component	Weight	CLO Focus	Mode
Online Test 1 (Ch 1–4)	10%	CLO1	Timed online quiz, auto-graded
Online Test 2 (Ch 5–6)	10%	CLO1 + CLO2	Timed online quiz, auto-graded
Performance Analysis Assignment	20%	CLO2	Individual written report
System-Level Mini Project	20%	CLO3	Individual code + recorded demo
Continuous Assessment Total	60%	—	—
Final Case-Study Examination	30%	CLO1 + CLO2 + CLO3	Timed open-book online exam
Individual Demonstration	10%	CLO3	1-to-1 synchronous video call
Final Assessment Total	40%	—	—
Grand Total	100%	—	—

FINAL GRADE

PLEASE NOTE:
To pass this module, it is required to pass both assessment components. Only then will the final grade be considered as PASS



Assessment Component	Requirement to Pass	Weightage	Contribution to Overall Mark
Continuous Assessment	Minimum 30%	60%	30%
Final Assessment	Minimum 40/100	40%	16%
Overall Passing Requirement	Must pass both components	100%	Minimum 46% combined



Rules Regulations

course completion	Make sure submitting all class activities and assessments
Assessments	All submissions must be the student's own work. Plagiarism will result in disciplinary action.
Deadlines	All deadlines are final. Extensions will only be granted in exceptional conditions and must be requested in advance.



Attendance requirements

Students are required to attend at least 80% of classes in a semester.

warning letter:
If a student misses 2 consecutive classes without proper justification

Code of Conduct



- Respectful Communication: All communication within the any environment must be respectful and professional.

- Academic Integrity: Students must adhere to the highest standards of academic integrity. Cheating, plagiarism, and any form of dishonesty will result in severe penalties.

Lecturer Information

Ms Siti Nazirah Bt Muhamad Zainuldin

sitinazirahzainuldin@segi.edu.my

Office Venue: Level 2 (Main Campus SEGi Kota Damansara)

Consultation Hours: Email for meeting

THANKS